High-reliability sites in V1 and V2 for sample separation

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1 Distance between samples

Based on `20260909`, the first pass results will be based on the pairwise sample distances, reported as the Mahalanobis distance with the Ledoit-Wolf shrinkage covariance estimator. The family-wise averages of the randomized dichotomy distances and the pairwise sample distances seem to be well correlated, so the overall results should be similarly reliable.

2 Reliable site distributions

Figure 1 is a copy of the empirical CDFs and histograms of the split-half correlations for the sites, originally calculated in `20260826`. Previous results looked at the results for sites with correlations between 0.3 and 0.6.

Split-half correlations in V1 and IT are similarly shaped, with a right-skewed distribution and a mode around $\rho \approx 0.1$. For V2, the correlations are left-skewed. V4 has a somewhat bimodal distribution, with a local peak around $\rho \approx 0.1$ and $\rho \approx 0.6$.

3 Pairwise samples compared with randomized dichotomies

Pairwise sample comparisons are done separately for each family.

Visually, the distributions between the pairwise sample comparisons and the randomized dichotomies look similar, though the magnitudes of the distances are affected by the different number of samples (Figure 2). The orderings on family-separated results are also similar (these plots are saved but not rendered here). When plotting the within-family averages and standard deviations against each other (Figure 3), the correlations tend to be good for within-family averages and worse for within-family standard deviations (Table 1).

To make this more rigorous, one could test for similarity between the relative rankings of region performance with a post hoc test. However, this is probably overkill for now.

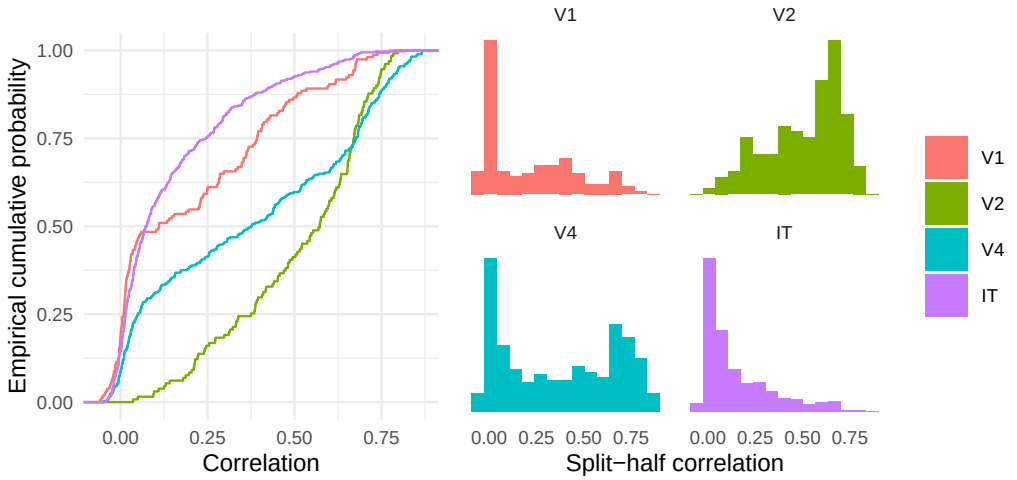


Figure 1: Empirical cumulative distribution functions (left) and histograms (right) for the split-half correlations by region. Data reflects split-half correlations performed when all blanks are averaged. V1 and IT are right-skewed, V2 is left-skewed, and V4 has a somewhat bimodal distribution for low- and high-correlations.

Table 1: The correlations between family-wise summary statistics of mean and standard deviation for each region and in naturalistic or noisy families. The correlations of family-wise means tends to be high for both types of texture families. Correlations in within-family SD tend to drop in noisy families.

	Naturalistic		Noisy	
Region	Corr(Means)	Corr(SD)	Corr(Means)	Corr(SD)
V1	0.929	0.785	0.892	0.590
V2	0.922	0.794	0.898	0.682
V4	0.919	0.344	0.908	0.357
IT	0.957	0.578	0.963	0.281

I am fairly unworried about big differences between the results from the pairwise sample distances and the randomized dichotomies. Subsequent analyses will be performed on the pairwise sample distances, which will save time (and memory).

4 References

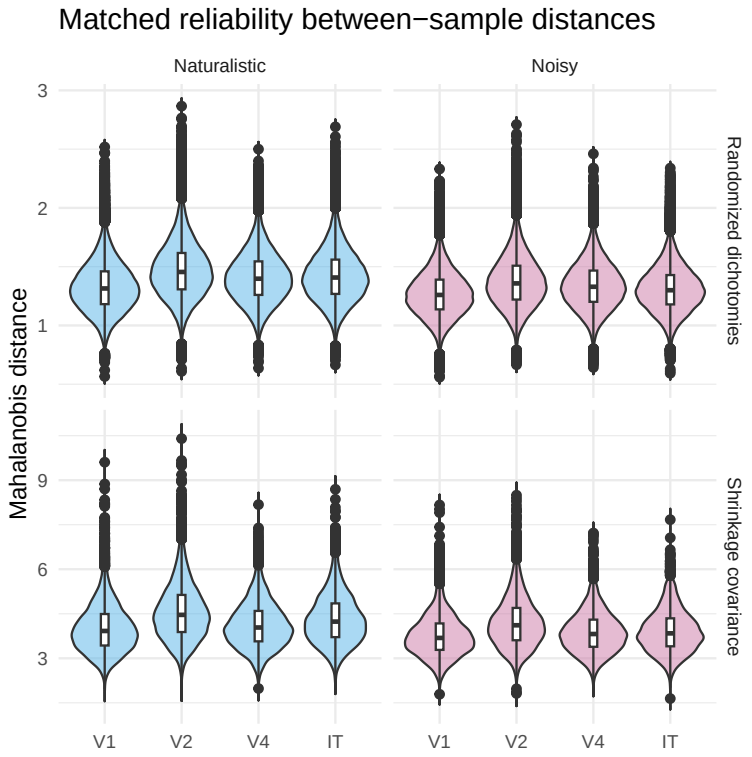


Figure 2: Violin plots comparing the distributions of Mahalanobis distances for the randomized dichotomies approach (top) and the pairwise between-sample distances found with the Ledoit-Wolf shrinkage estimator.

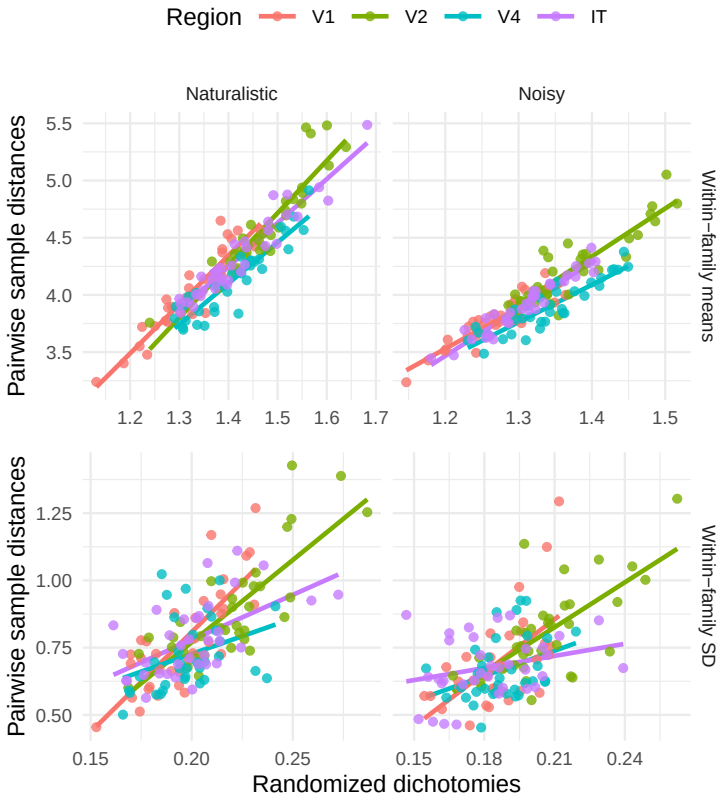


Figure 3: Comparison of summary statistics of the distances found with pairwise sample distances and randomized dichotomies. Means and standard deviations are calculated within each texture family. The within-family means tend to show more correlation across all regions. The correlation between the pairwise sample and randomized dichotomies standard deviations is worse in IT than other regions.